Problem 10.9

At time 0, K is deposited into Fund X, which accumulates at a force of interest

$$\delta_t = 0.006t^2.$$

At time m, $2K$ is deposited into Fund Y, which accumulates at an annual effective interest rate of 10%. At time n, where $n > m$, the accumulated value of each fund is $4K$. Determine m.

Solution

For Fund X, the accumulated value at time n is

$$K \exp\left(\int_0^n 0.006t^2\, dt\right).$$

Since this value equals $4K$,

$$4K = K \exp\left(\int_0^n 0.006t^2\, dt\right).$$

Canceling K and evaluating the integral gives

$$4 = \exp\left(0.006\frac{n^3}{3}\right) = e^{0.002n^3}.$$

Taking natural logarithms,

$$\ln 4 = 0.002n^3.$$

Therefore,

$$n = \left(\frac{\ln 4}{0.002}\right)^{1/3} = 8.849970445.$$

For Fund Y, the deposit of $2K$ accumulates from time m to time n at an annual effective rate of 10%. Thus,

$$4K = 2K(1.10)^{n-m}.$$

Canceling $2K$ gives

$$2 = (1.10)^{n-m}.$$

Taking natural logarithms,

$$n - m = \frac{\ln 2}{\ln 1.10}.$$

Hence,

$$m = n - \frac{\ln 2}{\ln 1.10}.$$

Substituting the value of n,

$$m = \left(\frac{\ln 4}{0.002}\right)^{1/3} - \frac{\ln 2}{\ln 1.10}.$$

Therefore,

$$\boxed{m = 1.577429548}.$$